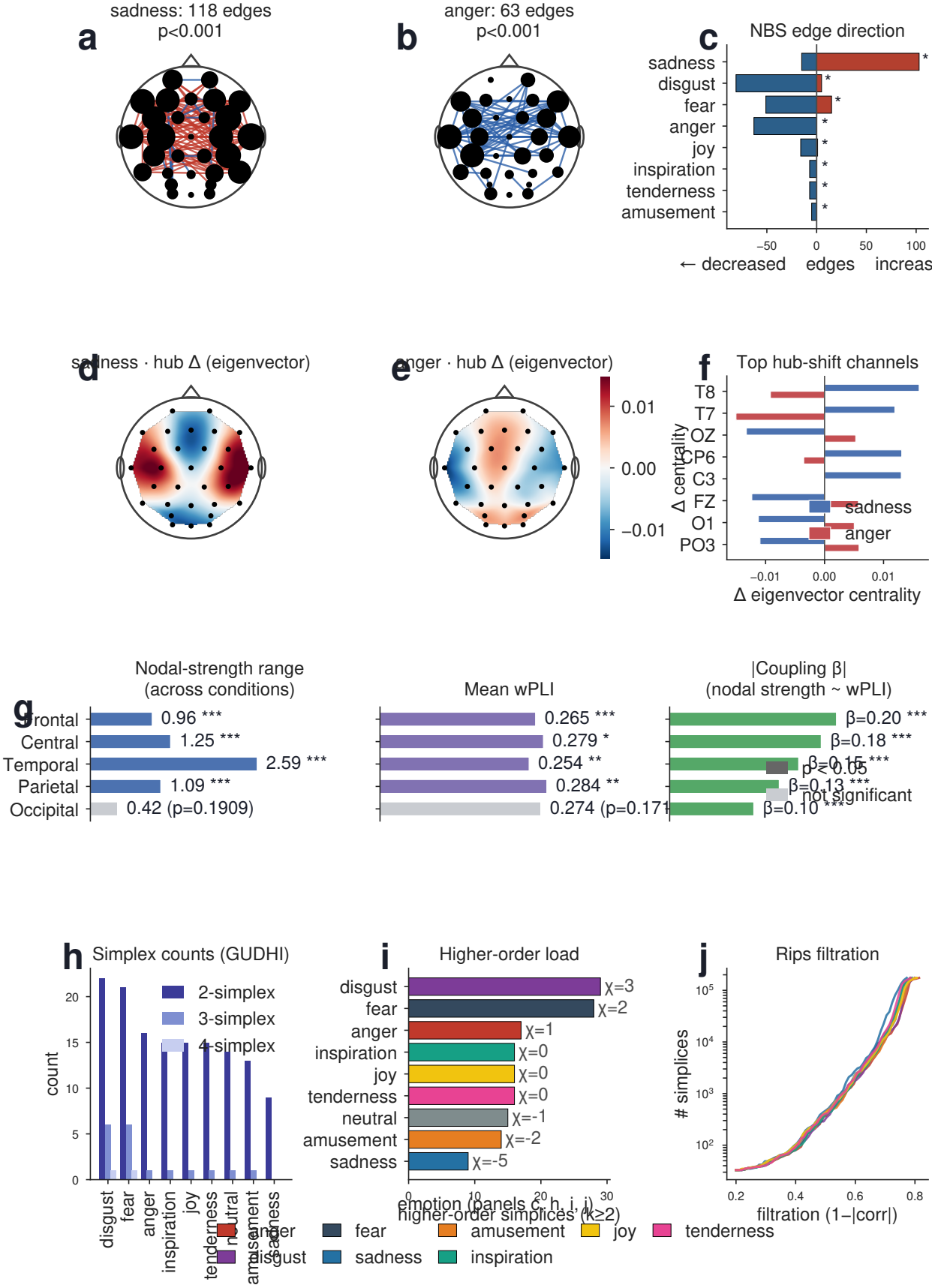

a
sadness: 118 edges
p<0.001
b
anger: 63 edges
p<0.001
c
NBS edge direction
sadness
disgust
fear
anger
joy
inspiration
tenderness
amusement
−50
0
50
100
← decreased edges increas
d
sadness · hub Δ (eigenvector)
e
anger · hub Δ (eigenvector)
0.01
0.00
−0.01
f
Top hub-shift channels
T8
T7
OZ
CP6
C3
FZ
O1
PO3
Δ centrality
sadness
anger
−0.01
0.00
0.01
Δ eigenvector centrality
g
Nodal-strength range
(across conditions)
Frontal 0.96 ***
Central 1.25 ***
Temporal 2.59 ***
Parietal 1.09 ***
Occipital 0.42 (p=0.1909)
Mean wPLI
0.265 ***
0.279 *
0.254 **
0.284 **
0.274 (p=0.171)
|Coupling β|
(nodal strength ~ wPLI)
β=0.20 ***
β=0.18 ***
β=0.15 ***
β=0.13 ***
β=0.10 ***
p < 0.05
not significant
h
Simplex counts (GUDHI)
2-simplex
3-simplex
4-simplex
count
0
5
10
15
20
disgust
fear
anger
inspiration
joy
tenderness
neutral
amusement
sadness
i
Higher-order load
disgust χ=3
fear χ=2
anger χ=1
inspiration χ=0
joy χ=0
tenderness χ=0
neutral χ=-1
amusement χ=-2
sadness χ=-5
0
10
20
30
higher-order simplices (k≥2)
j
Rips filtration
simplices
10^2
10^3
10^4
10^5
0.2
0.4
0.6
0.8
filtration (1−|corr|)
emotion (panels c, h, i, j)
anger
fear
amusement
joy
tenderness
disgust
sadness
inspiration